



r/LocalLLaMA • 1 yr. ago

rag_perplexity



Implementing function calling (tools) without frameworks?

Discussion

Generally it's pretty doable (and sometimes simpler) to write whole workloads without touching a framework. I find calling the component's APIs and just straight python works easier a lot of time than twist the workloads to fit someone else's thinking process.

I'm ok with using some frameworks to implement agentic workflows with tools/functions but wondering if anyone here just implemented it with just old fashioned coding using local llms. This is more of a learning exercise than trying to solve a problem.



9



22



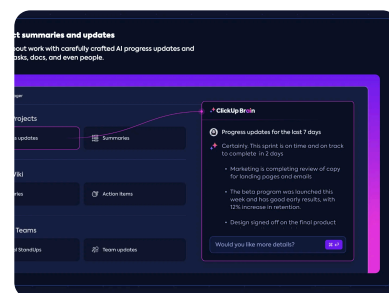
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Such_Advantage_6949 • 1y ago

Yes, i had just decided to give up on langchain and langgraph (after like 10 tries). Ultimately coding something myself seems easier. Granted it might not have as much feature but at least i know where i can tweak thing. I will leverage on function from those framework where convenient e.g. rag, tools. But for the agent orchestration, i am building my own which is kinda similar to langgraph but i dont need to touch the bloody LCEL and runnable thingy.



7

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5 more replies



kryptkpr • 1y ago

[Hermes-Function-Calling](#)

6

[Reply](#)[boris_and_proud](#) • 1y ago

even though the llm has produced the correct output :(

↑ 2 ↓  Reply ...



Open_Channel_8626 · 1y ago

Frameworks don't really do anything to solve the main problem, which is the LLM being able to use the right tools and the right parameters at the right time.

I am eagerly waiting for GPT 5 (and then one more year wait for GPT 5 level open source models) because I don't think GPT 4 level models or below are reliable yet at tool use.

↑ 7 ↓  Reply ...

 3 more replies



mrjackspade · 1y ago

I've got the model calling the functions at the right times and using the proper syntax (8x22B) but I haven't wired up the response yet. I've got the model using the format `@invoke("function_name", parameters)`

Honestly past that point its pretty trivial. Its just a matter of actually executing the function.

Since I'm working in C# that really only involves using reflection to find and invoke the function, and converting the parameters to the proper data type.

↑ 3 ↓  Reply ...



segmond · 1y ago

 Top 1% Commenter

Yes, your code passes the function definition/tool specs to the LLM, your code passes the user input to the LLM, your code captures the output from the LLM, you inspect the output to see if the LLM decided to call a tool, if it did, you extract the function and parameters, You call the function with the given parameters, you take the result. You pass the result back to the LLM, the LLM combines the result with it's output, you present the output to the user.

 ↑ 4 ↓  Reply ...



Noiselexer · 1y ago

Indeed. So might as well use a framework that does all that... Lol

↑ 2 ↓  Reply ...

 1 more reply



remyxai · 1y ago

[FFMPerative](#) is an example using a fine-tuned 3B model and an abstract syntax tree. llama.cpp and llamafile helped to simplify dependencies in packaging.

↑ 1 ↓  Reply ...

did implement various parsers/mini frameworks for various local models like gorilla-llm/gorilla-openfunctions-v2, cognitivecomputations/fc-dolphin-2.6-mistral-7b-dpo-laser and some self trained ones lately. most of the time it didnt take more then 40 lines of code to implement the whole logic and maybe 60 - 100 more lines for various tools like wolfram search, calculator, web search, summarization, weather etc. and the good thing on it is that the execution is 5-100 times faster then when using a framework with a model thats not properly trained for the synthax the framework is expecting and then the back and forth starts.

↑ 1 ↓  Reply ...



tensorwar9000 · 1y ago

unfortunately, you need a framework to do the intermediary calls to the localLLM, and constrain generation (e.g. with regex and CFGs). This, and they all suck. One that I haven't tested that looks promising was function tool calling with LocalAI

<https://github.com/mudler/LocalAI/tree/master/examples/functions>

I have been meaning to check it out but haven't. What I need is near or exact to assistant api.

↑ 0 ↓  Reply ...



MasterDragon_ · 1y ago



This before is an example taken from open ai docs. This is all you need to implement function calling

```
from openai import OpenAI
```

```
client = OpenAI()
```

Example dummy function hard coded to return the same weather

In production, this could be your backend API or an external API

```
def get_current_weather(location, unit="fahrenheit"): """Get the current weather in a given location""" if "tokyo" in location.lower(): return json.dumps({"location": "Tokyo", "temperature": "10", "unit": unit}) elif "san francisco" in location.lower(): return json.dumps({"location": "San Francisco", "temperature": "72", "unit": unit}) elif "paris" in location.lower(): return json.dumps({"location": "Paris", "temperature": "22", "unit": unit}) else: return json.dumps({"location": location, "temperature": "unknown"})
```

```
def run_conversation(): # Step 1: send the conversation and available functions to the model messages = [{"role": "user", "content": "What's the weather like in San Francisco, Tokyo, and Paris?"}] tools = [ { "type": "function", "function": { "name": "get_current_weather", "description": "Get the current weather in a given location", "parameters": { "type": "object", "properties": { "location": { "type": "string", "description": "The city and state, e.g. San Francisco, CA", }, "unit": { "type": "string", "enum": ["celsius", "fahrenheit"] }, }, "required": ["location"], }, }, } ] response = client.chat.completions.create( model="gpt-4o", messages=messages, tools=tools, tool_choice="auto", # auto is default, but we'll be explicit ) response_message = response.choices[0].message tool_calls = response_message.tool_calls # Step 2: check if the model wanted to call a function if tool_calls: # Step 3: call the function # Note: the JSON response may not always be valid; be sure to handle errors available_functions = { "get_current_weather": get_current_weather, } # only one function in this example, but you can have multiple messages.append(response_message) # extend conversation with assistant's reply # Step 4: send the info for each function call and function response to the model for tool_call in tool_calls: function_name = tool_call.function.name function_to_call = available_functions[function_name] function_args = json.loads(tool_call.function.arguments) function_response = function_to_call( location=function_args.get("location"), unit=function_args.get("unit"), ) messages.append( { "tool_call_id": tool_call.id, "role": "tool", "name": function_name, "content": function_response, } ) # extend conversation with function response second_response = client.chat.completions.create( model="gpt-4o", messages=messages, ) # get a new response from the model where it can see the function response return second_response print(run_conversation())
```

⊖ ↑ -1 ↓ 🗨 Reply ...



rag_perplexity OP • 1y ago

Sorry should of specified local LLMs. For OpenAI I'm assuming the python/tool execution is handled on their end?

↑ -1 ↓ 🗨 Reply ...

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How do I properly use function calling on Ollama models?

8 upvotes · 7 comments



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5 upvotes · 15 comments



r/LocalLLaMA • 3 mo. ago

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16 upvotes · 13 comments



r/LocalLLaMA • 7 mo. ago

Best Function Calling LLMs

16 upvotes · 12 comments



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10 upvotes · 24 comments



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19 upvotes · 20 comments



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Best open source LLM for function calling

18 upvotes · 16 comments



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Are there any use cases for text (non-instruct) models?

6 upvotes · 8 comments



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What are your use cases for small (1-3-8B) models?

91 upvotes · 116 comments



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69 upvotes · 10 comments



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What are your favorite local models for autocompletion using continuedev?

5 upvotes · 3 comments



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3 upvotes · 3 comments



r/OpenAI • 1 yr. ago

Any developers building complex web apps with OpenAI and nested JSON AI responses?

Experiences with running Python scripts in Power BI

1 upvote · 2 comments

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Devs using laravel + inertia js (React + typescript), how do you guys setup tests for frontend components?

3 upvotes

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Any usable ~7B models that actually work for function calling?

28 upvotes · 25 comments

 r/LocalLLaMA · 3 mo. ago

Function calling in llama.cpp?

10 upvotes · 5 comments

 r/LocalLLaMA · 9 mo. ago

Looking for a simple practical example of using a function call with a Ollama model.

3 upvotes · 8 comments

 r/LocalLLaMA · 4 mo. ago

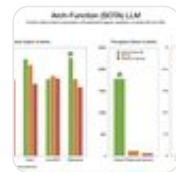
Function Calling

7 upvotes · 15 comments

 r/LocalLLaMA · 3 mo. ago

I built a small (function calling) LLM that packs a big punch; integrated in an open source gateway for agentic apps

217 upvotes · 72 comments



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